

Multi-AS BGP Network Simulation

with Traffic Engineering

Personal Project Notes | April 26, 2026

Topology

5-router, 3-AS topology simulating an ISP core (AS100) peering with a customer (AS200) and a peer ISP (AS300). Built and verified in GNS3 using FRRouting.

```
AS 100 (ISP Core)
+-----+
| R1 Route |
| Reflector |
+-----+
| iBGP |
+-----+ +-----+
| R2 | | R3 |
| ASBR1 | | ASBR2 |
+-----+ +-----+
| eBGP | eBGP
+-----+ +-----+
| AS200 | | AS300 |
| R4 | | R5 |
+-----+ +-----+
```

AS Design

AS	Role	Routers	Loopbacks
100	ISP Core	R1 (RR), R2 (ASBR1), R3 (ASBR2)	10.0.0.x/32
200	Customer	R4	192.168.200.0/24
300	Peer ISP	R5	192.168.300.0/24

IP Addressing

Link	Subnet	R1/R2 End	R2/R3/R4/R5 End
R1–R2 (iBGP)	10.1.12.0/30	10.1.12.1	10.1.12.2
R1–R3 (iBGP)	10.1.13.0/30	10.1.13.1	10.1.13.2
R2–R4 (eBGP)	172.16.24.0/30	172.16.24.1	172.16.24.2
R3–R5 (eBGP)	172.16.35.0/30	172.16.35.1	172.16.35.2

What Was Built & Verified

GNS3 Lab Environment

- GNS3 2.2.58.1 on Windows with GNS3 VM running in VirtualBox
- FRRouting (FRR) v8.2.2 as the router OS — full BGP/OSPF support
- 5 routers connected via virtual links matching the topology above

BGP Features Configured

- iBGP Route Reflector (R1) — eliminates full-mesh requirement inside AS100
- OSPF IGP inside AS100 for loopback reachability (iBGP uses loopbacks)
- eBGP sessions: R2-R4 (AS100-AS200) and R3-R5 (AS100-AS300)
- LOCAL_PREF 150 for customer routes (AS200) — preferred over peer
- LOCAL_PREF 100 for peer routes (AS300) — standard peer preference
- Prefix-list filtering on all eBGP sessions — no route leaking
- Route-maps for inbound/outbound policy on all ASBR interfaces

Verification Results

Confirmed on R1 (Route Reflector) after full topology came up:

```
R1-AS100-RR# show bgp summary
Neighbor AS MsgRcvd MsgSent Up/Down PfxRcd
10.0.0.2 100 8 9 00:03:49 1
10.0.0.3 100 5 5 00:02:18 0

R1-AS100-RR# show ip bgp
Network Next Hop LocPrf Path
10.0.0.0/16 0.0.0.0 32768 i
192.168.200.0/24 172.16.24.2 150 0 200 i
```

- Both iBGP neighbors Established (numeric PfxRcd = Established in FRR)
- 192.168.200.0/24 (AS200 Customer) visible at R1 with LOCAL_PREF 150
- OSPF routes for all loopbacks and links present in routing table

Python Automation (Netmiko)

- bgp_audit.py — concurrent BGP health check across all 5 routers
- push_config.py — config deployment with dry-run mode, group targeting
- collect_routes.py — BGP route snapshots + diff between snapshots

Next Steps

- Simulate link failure on R2-eth1, observe BGP reconvergence
- Add BGP communities for advanced traffic engineering
- Run Netmiko scripts against live lab
- Push to GitHub with README for AT&T; portfolio

